Real-Time Flight Delay Prediction

MIS 584 Team NYJ

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Problem

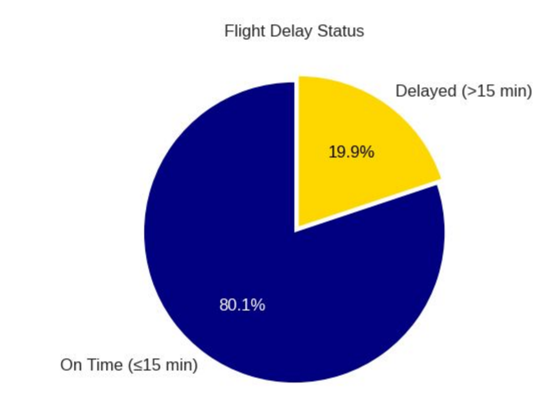
- Flights are frequently delayed from scheduled times
- Provide real-time delay predictions
  - Airlines’ arrival predictions, last-minute travel decisions, final destination transportation
- Identify common factors in delayed flights
  - Creates flight recommendations to help passengers choose airlines, travel dates, and layover locations
  - Potential to inform original flight schedules
- Approach: classification → regression

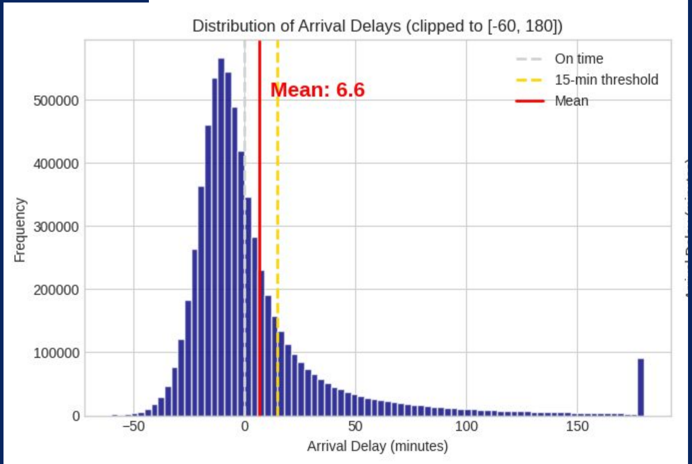
Data

- The data being used is an open source, publicly available dataset on Kaggle "US 2023 Civil Flights, delays, meteo and aircrafts".
- Does not need to fulfill any IRB requirements.
- 5 CSV Files:
  - **US_flights_2023.csv** — Core dataset: flight characteristics, delay durations, airline, aircraft info (manufacturer, model, age).
  - **cancelled_diverted.csv** — Separated to avoid bias in delay analysis.
  - **airports_geolocation.csv** — Airport geographic data (lat, long, city, state).
  - **weather_meteo_by_airport.csv** — Daily weather per airport; joinable to main table via airport_id.
  - **maj_us_flight_january_2024.csv** — Redundant data

Trends and Flight Recommendations

- 1 in 5 flights is delayed (>15 min); average arrival delay is 6.6 min across 6.7M flights
- Afternoon/evening departures, Fridays/Sundays, and June–July are the highest-risk combinations
- JetBlue & Frontier consistently underperform; Republic & Endeavor Air arrive ahead of schedule
- Heavy precipitation and storm-force winds are strong delay triggers — extreme rain alone doubles delay probability (17% → 38%)





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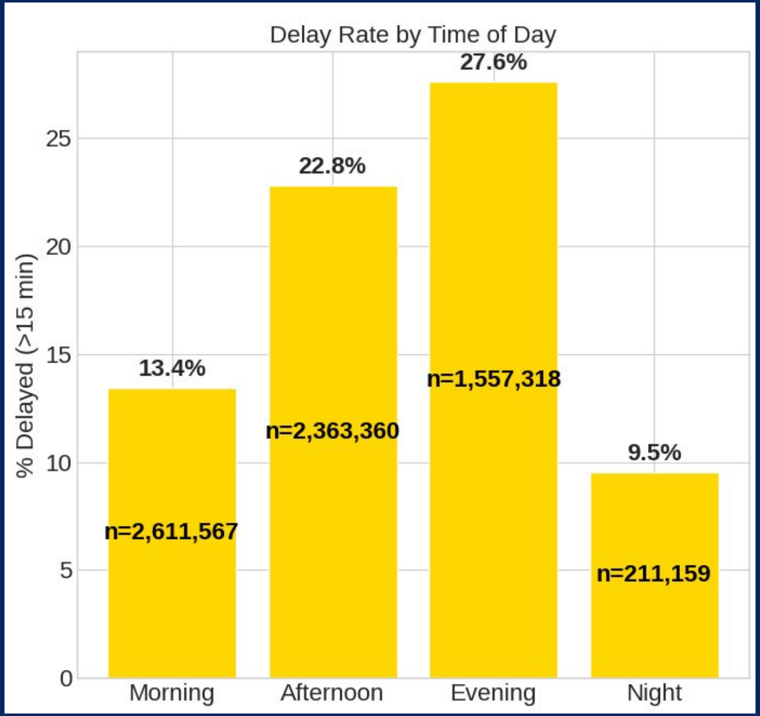

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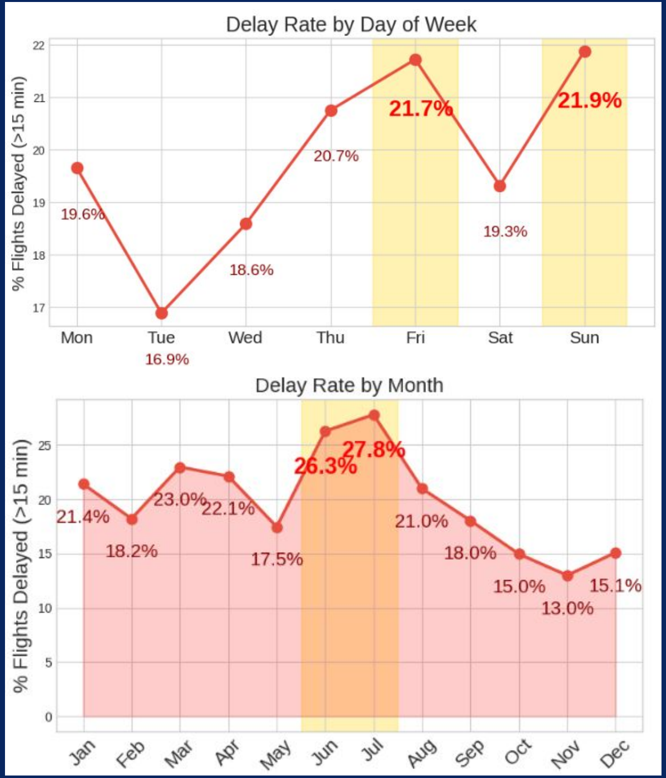

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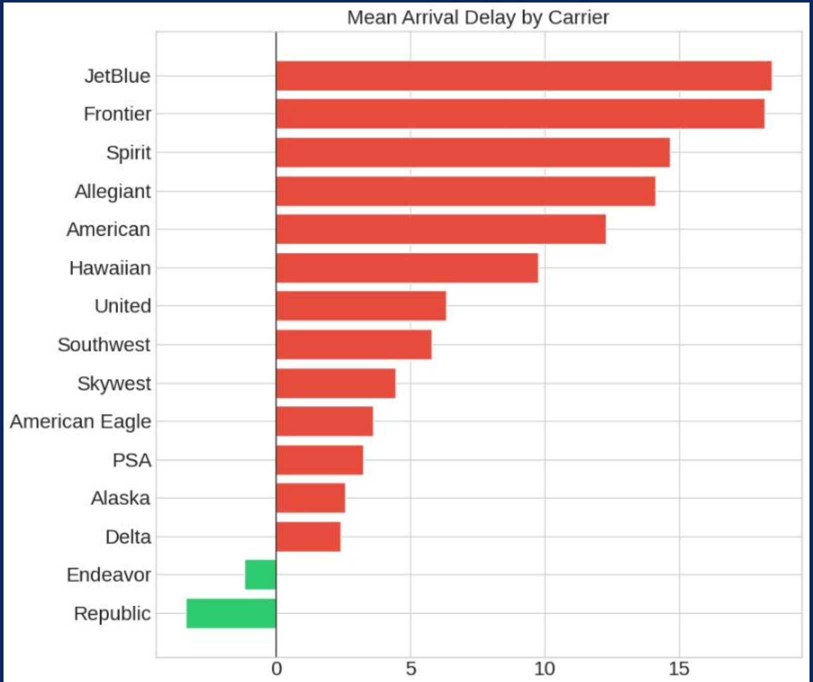

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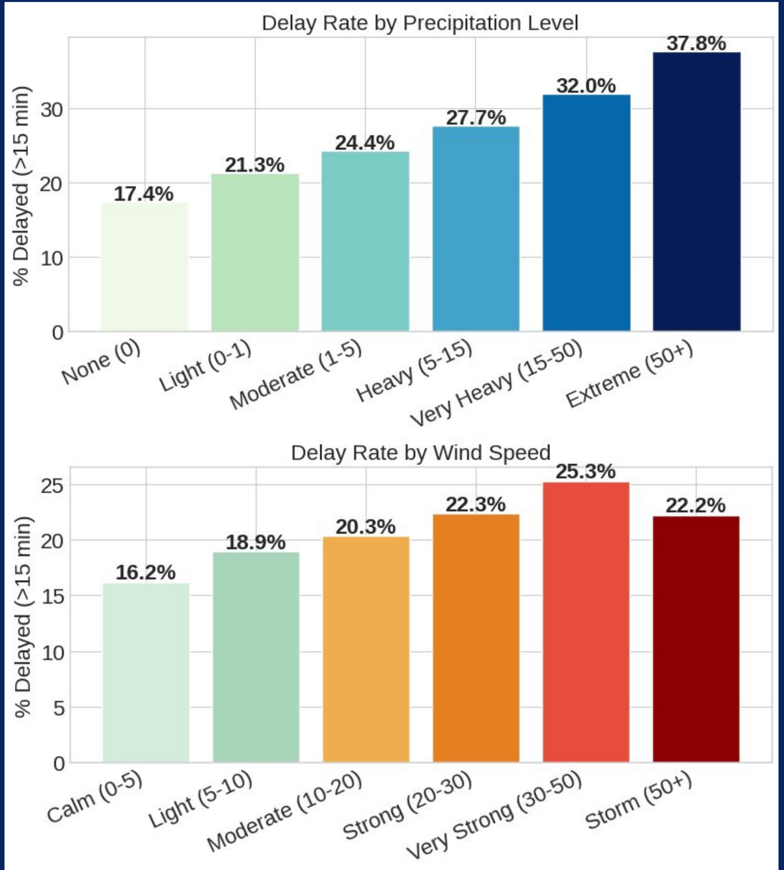
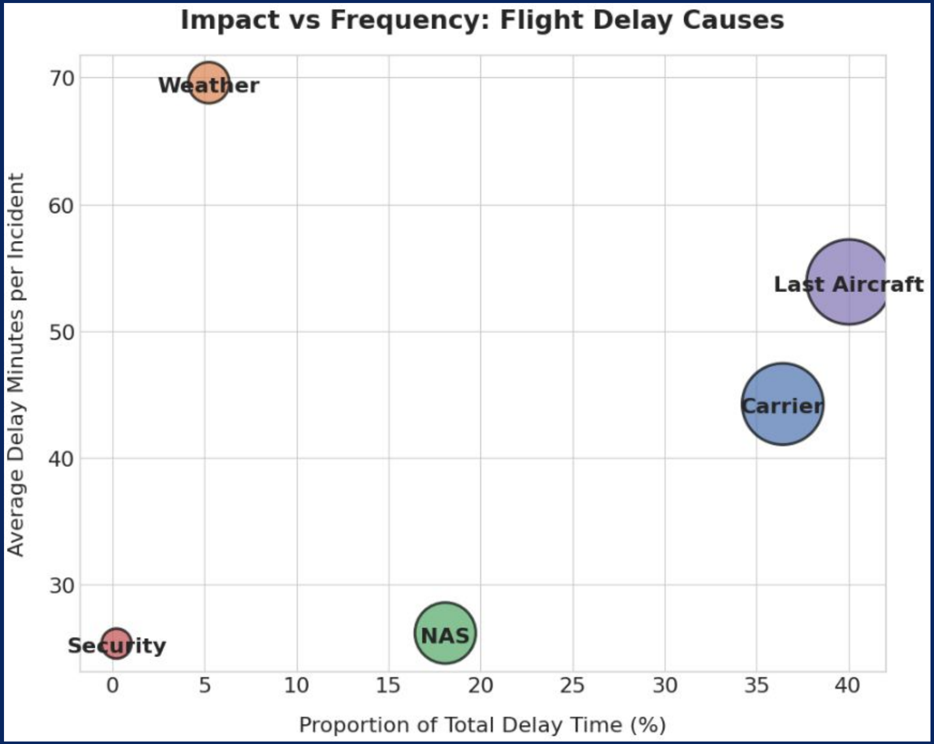


Trends and Flight Recommendations

- Last Aircraft & Carrier delays dominate total delay time (~75% combined); Weather has highest per-incident impact but low frequency
- Florida airports cluster at the top of delay rankings — FLL (30.4%), MCO (27.9%), MIA (26.5%) form the worst-performing region
- Mid-sized airports in the Midwest & Mountain West dominate punctuality — GSP (13.2%), GSO (13.8%), TUL (14.0%) lead with delay rates roughly half those of Florida hubs



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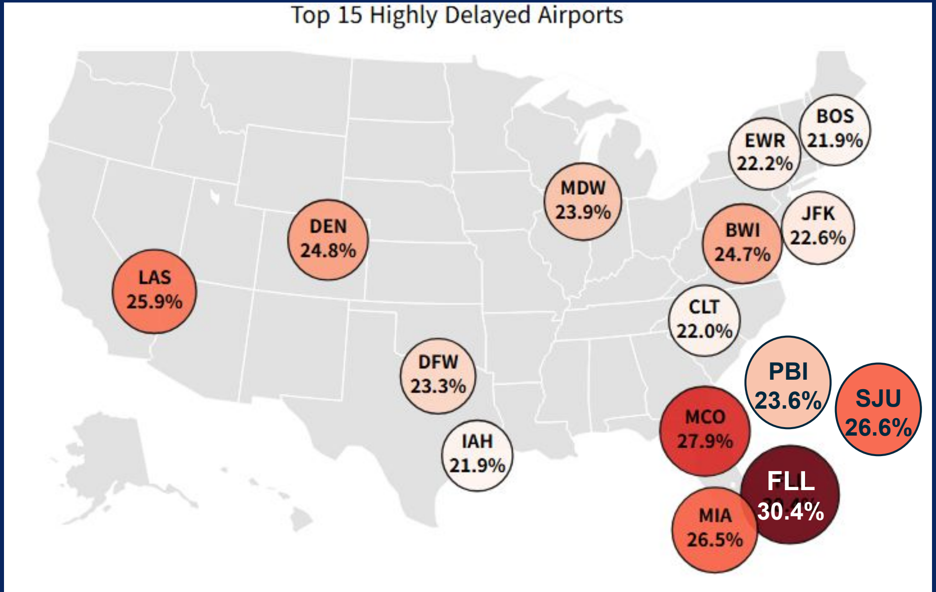

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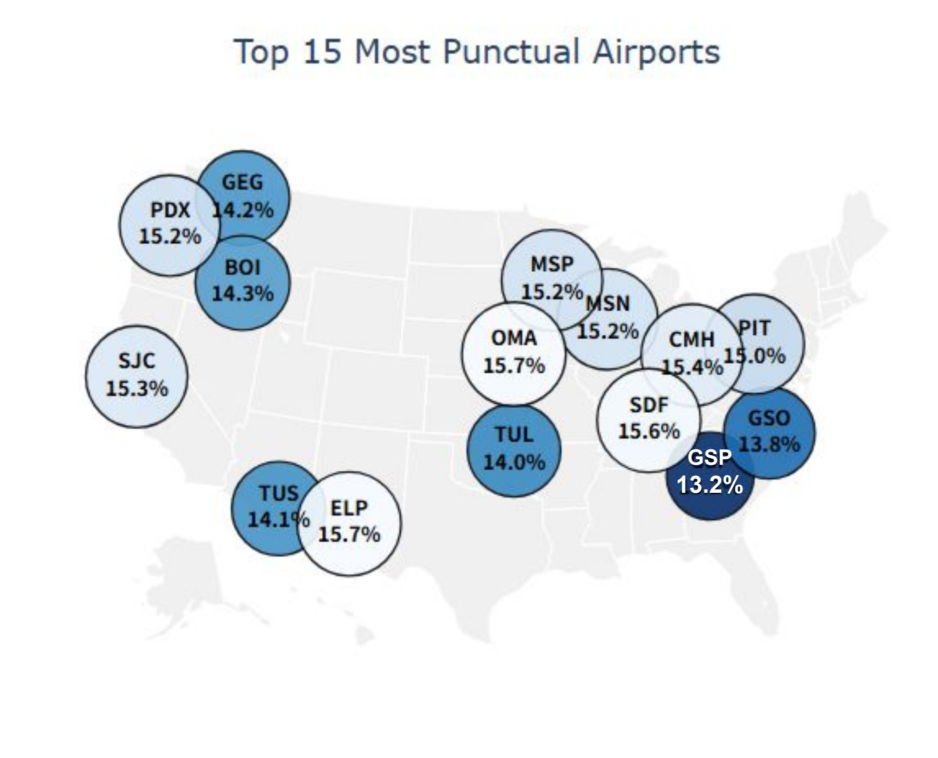


Data Preparation and Feature Engineering

Data Preparation and Cleaning

- Joined flights and weather datasets
- Removed outliers duplicate rows
- Impute null values of dealy and weather cols (snow, prcp) to 0.
- Impute cols like aircraft age with mean.

Feature Engineering and Selection

- Extracted date features
- Dropped redundant, uninformative, and highly correlated features
- Applied one-hot encoding for unordered categorical variables

Model Preparation

- Standardized features for distance-based models
- Created same 80/20 train-test split for all models
- Filtered dataset for delayed flights for regression (where, flight delayed = 1)

Classification Algorithms

Classification Task:

- Will the flight be delayed or not?

Models:

- Decision Tree - shows which factors most influence delay.
- Random Forest - ensemble of trees, more accurate and reliable predictions.
- Logistic Regression - easy to interpret binary classifier.
- Support Vector Machine - strong generalization and less prone to overfitting.

Classification Results

Test Results with all features

Model	AUC_ROC	Recall	Accuracy	F1
Decision Tree	0.9999	0.9977	0.9977	0.9977
Linear SVM	0.9995	0.9349	0.9349	0.9298
Logistic Regression	0.9995	0.9635	0.9635	0.962
Random Forest	0.9995	0.9821	0.9821	0.9819

Classification Results

Test Results with no delay features
(Dep_Delay, Delay_Carrier, Delay_Weather, Delay_NAS, Delay_Security, Delay_LastAircraft)

Model	AUC_ROC	Recall	Accuracy	F1
Decision Tree	0.5336	0.8053	0.8053	0.7339
Linear SVM	0.6204	0.8022	0.8022	0.7142
Logistic Regression	0.6659	0.8021	0.8021	0.7202
Random Forest	0.6655	0.8022	0.8022	0.7142

Regression Algorithms

Regression Task:

- Given a flight will be delayed, predict how long it will be delayed (in minutes).

Models:

- Decision Tree - shows which factors most influence delay length.
- Random Forest - ensemble of trees, more accurate and reliable predictions.
- Gradient Boosted Trees - highest performing model, captures complex patterns for real-time delay prediction.
- Linear Regression - simple baseline for predicting delay duration.

Regression - Decision Tree

	RMSE (minutes)	MAE (minutes)	R^2
Train	39.94	10.56	0.8302
Test	42.40	10.88	0.8079

- No overfitting observed
- Reasonable predictions on average
- Some extreme predictions

Regression Results

Model	RMSE (minutes)	MAE (minutes)	R^2
Decision Tree	42.40	10.88	0.8079
Random Forest	42.10	12.69	0.8107
Gradient Boosted Trees	41.38	8.59	0.8164
Linear Regression	0.63	0.45	1.0000

Regression Results

Removed Dep_Delay, Delay_Carrier, Delay_Weather, Delay_NAS, Delay_Security, Delay_LastAircraft

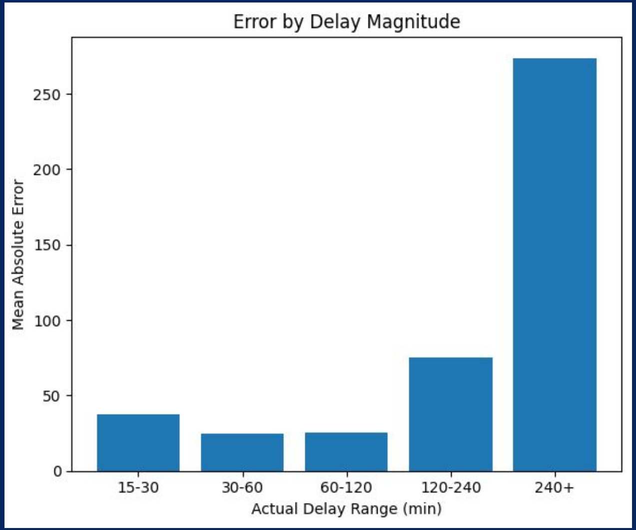
Model	RMSE (minutes)	MAE (minutes)	R^2
Decision Tree	95.50	49.81	0.0238
Random Forest	94.72	49.67	0.0362
Gradient Boosted Trees	94.74	49.23	0.0356
Linear Regression	95.03	50.07	0.0297

★ Exact delay predictions highly dependent on day-of delay information

Regression with Iterative Improvement

- Baseline : LightGBM
- Feature Engineering : cyclical encoding + target encoding + binary indicators
- Fine Tuning : Optuna's Bayesian optimization across 30 trials
- Information Ceiling (R^2=0.26): Failure to capture variance leads to regression to the mean and systemic errors at delay extremes

Stage	RMSE (minutes)	MAE (minutes)	R^2
Baseline	48.06	91.53	0.105
Feature Engineering	48.08	91.49	0.106
Fine Tuning	44.02	83.13	0.262



Final System | Two-Stage Prediction

Key Insight — Extreme delays (>143 minutes, top 10%) are inherently unpredictable events. Isolating these outliers allows the model to focus on the normal range

Input Data **266,700**

Tier 1: Classifier
Severe Delay(>143 minutes)

NO 190,473 (71.4%)

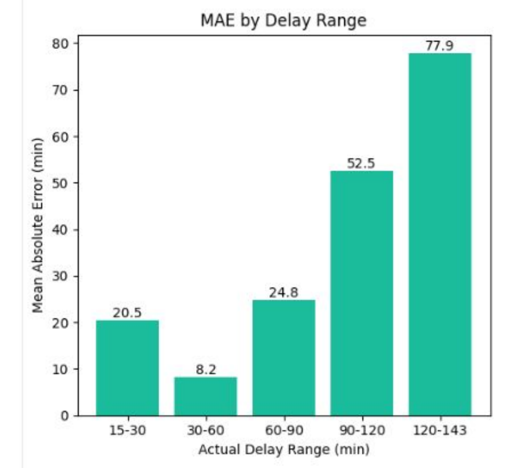
Tier 2:Regression
Provides specific predictions for "X-minute delays" (ranging from 15 to 143 minutes).

YES 76,227 (28.6%)

Anomaly Alert

"Severe Delay" Risk Warning Issued (Unpredictable Case)

Model	RMSE (minutes)	MAE (minutes)	R^2
Normal range	20.75	26.70	0.149



Conclusion

- Trends: Avoid severe weather, weekends, summer, afternoons/evenings, JetBlue & Frontier airlines, and Florida airports
- Classification: Random forest performs best
  - High AUC and strong recall and accuracy
  - Models nonlinear relationships
- Regression: Linear regression achieves the best results when using all features
  - Day-of delay information needed for accurate predictions
- Regression Improvement: Two-tier architecture with outlier separation reduced MAE from 44 to 20.75 minutes on normal-range flights

Thank You